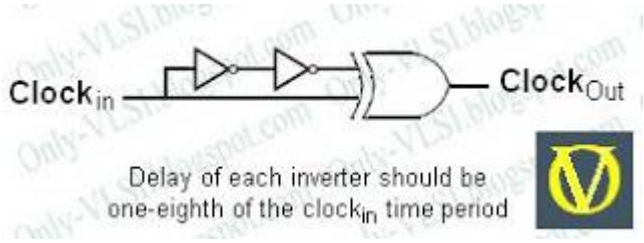
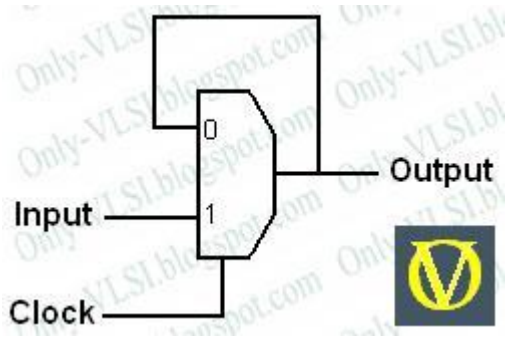
2. Design a circuit which doubles the frequency of a given input clock signal.
Answer



3. Implement a D-latch using 2x1 multiplexer(s).
Answer



3. What are various types of state encoding techniques? Explain them.
Answer

One-Hot encoding: Each state is represented by a bit flip-flop). If there are four states then it requires four bits (four flip-flops) to represent the current state. The valid state values are 1000, 0100, 0010, and 0001. If the value is 0100, then it means second state is the current state.

One-Cold encoding: Same as one-hot encoding except that '0' is the valid value. If there are four states then it requires four bits (four flip-flops) to represent the current state. The valid state values are 0111, 1011, 1101, and 1110.

Binary encoding: Each state is represented by a binary code. A FSM having '2 power N' states requires only N flip-flops.

Gray encoding: Each state is represented by a Gray code. A FSM having '2 power N' states requires only N flip-flops.

10. Define Clock Skew , Negative Clock Skew, Positive Clock Skew.
Answer

Clock skew is a phenomenon in synchronous circuits in which the clock signal (sent from the clock circuit) arrives at different components at different times. This can be caused by many different things, such as wire-interconnect length, temperature variations, variation in intermediate devices, capacitive coupling, material imperfections, and differences in input capacitance on the clock inputs of devices using the clock.
There are two types of clock skew: negative skew and positive skew. Positive skew occurs when the clock reaches the receiving register later than it reaches the register sending data to the receiving

register. Negative skew is the opposite: the receiving register gets the clock earlier than the sending register.

9. Compare and contrast between 1's complement and 2's complement notation.
Answer

The only advantage of 1's complement is that it can be calculated easily, just by changing 0's into 1's and 1's into 0's. The 2's complement is calculated in two ways, (i) add 1 to the 1's complement of the number, and (ii) leave all the leading 0s in the least significant positions and keep first 1 unchanged, and then change 0's into 1's and 1's into 0's.

The advantages of 2's complement over 1's complement are:
(i) For subtraction with complements, 2's complement requires only one addition operation, where as for 1's complement requires two addition operations if there is an end carry.
(ii) 1's complement has two arithmetic zeros, all 0's and all 1's.

1. What are the important aspects of VLSI optimization?
Answer

Power, Area, and Speed.

2. What are the sources of power dissipation?
Answer

+ Dynamic power consumption, due to logic transitions causing logic gates to charge/discharge load capacitance.
+ Short-circuit current, this occurs when p-tree and n-tree shorted (for a while) during logic transition.
+ Leakage current, this is a very important source of power dissipation in nano technology, it increases with decrease in lambda value. It is caused due to diode leakages around transistors and n-wells.

3. What is the need for power reduction?
Answer

Low power increases noise immunity, increases batter life, decreases cooling and packaging costs.

4. Give some low power design techniques.
Answer

Voltage scaling, transistor resizing, pipelining and parallelism, power management modes like standby modes, etc.

5. Give a disadvantage of voltage scaling technique for power reduction.
Answer

When voltage is scaled, designers tend to decrease threshold voltage to maintain good noise margins. But decreasing threshold voltages increases leakage currents exponentially.

6. Give an expression for switching power dissipation.
Answer

$P_{switching} = (1/2)CV_{dd}^2f$
Where
$P_{switching}$ = Switching power.
C = Load capacitance.
V_{dd} = Supply voltage.
f = Operating frequency.

7. Will glitches in a logic circuit cause power wastage?
Answer

Yes, because they cause unexpected transitions in logic gates.

8. What is the major source of power wastage in SRAM?
Answer

To read/write a word data, activates a word line for a row which causes all the columns in the row to be active even though we need only a word data. This consumes a lot power.

9. What is the major problem associated with caches w.r.t low power design? Give techniques to overcome it.
Answer

Cache is a very important part of the integrated chips, they occupy most of the space and hence contain lot of transistors. More transistors means more leakage current. That is the major problem associated with caches w.r.t. low power design. The following techniques are used to overcome it: V_{dd}-Gating, Cache decay, Drowsy caches, etc.

10. Does software play any role in low power design?
Answer

Yes, one can redesign a software to reduce power consumptions. For example modify the process algorithm which uses less number of computations.

1. Why is NAND gate preferred over NOR gate for fabrication?
Answer

NAND is a better gate for design than NOR because at the transistor level the mobility of electrons of NAND is normally three times that of holes compared to NOR and thus the NAND is a faster gate. The gate-leakage in NAND structures is much lower. If you consider t_phl and t_plh delays you will find that it is more symmetric in case of NAND (the delay profile), but for NOR, one delay is much higher than the other(obviously t_plh is higher since the higher resistance PMOSs are in series connection which again increases the resistance).

2. Which transistor has higher gain: BJT or MOSFET and why?
Answer

BJT has higher gain because it has higher transconductance.This is because the current in BJT is exponentially dependent on input where as in MOSFET it is square law.

3. Why PMOS and NMOS are sized equally in a transmission gates?
Answer

In transmission gate, PMOS and NMOS aid each other rather than competing with each other. So they are sized similarly.

4. What is SCR?
Answer

A silicon-controlled rectifier (or semiconductor-controlled rectifier) is a 4-layer solid state device that controls current flow.

An SCR is a type of rectifier, controlled by a logic gate signal. It is a 4-layered, 3-terminal device. A p-type layer acts as an anode and an n-type layer as a cathode; the p-type layer closer to the n-type(cathode) acts as a gate.

5. In CMOS digital design, why is the size of PMOS is generally higher than that of the NMOS?
Answer

In PMOS the carriers are holes whose mobility is less than the electrons, the carriers in NMOS. That means PMOS is slower than NMOS. In CMOS technology, NMOS helps in pulling down the output to ground and PMOS helps in pulling up the output to Vdd. If the sizes of PMOS and NMOS are the same, then PMOS takes long time to charge up the output node. If we have a larger PMOS than there will be more carriers to charge the node quickly and overcome the slow nature of PMOS. All this is done to get equal rise and fall times for the output node.

6. What is slack?
Answer

The slack is the time delay difference from the expected delay to the actual delay in a particular path. Slack can be positive or negative.

7. What is latch up?
Answer

A latchup is the inadvertent creation of a low-impedance path between the power supply rails of an electronic component, triggering a parasitic structure(The parasitic structure is usually equivalent to a thyristor or SCR), which then acts as a short circuit, disrupting proper functioning of the part. Depending on the circuits involved, the amount of current flow produced by this mechanism can be large enough to result in permanent destruction of the device due to electrical over stress - EOS.

8. Why is the size of inverters in buffer design gradually increased? Why not give the output of a circuit to one large inverter?
Answer

Because circuit can not drive the high output load straight away, so the load is gradually increased, by gradually increasing the size of inverters to get an optimized performance.

9. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus?
Answer

The charge sharing problem occurs when the charge which is stored at the output node in the phase is shared among the output or junction capacitances of transistors which are in the evaluation

phase. Charge sharing may degrade the output voltage level or even cause erroneous output value.

In the serially connected NMOS logic the input capacitance of each gate shares the charge with the load capacitance by which the logical levels drastically mismatched than that of the desired once. To eliminate this load capacitance must be very high compared to the input capacitance of the gate, which is generally 10 times.

10. What happens to delay if load capacitance is increased?
Answer

Delay increases.

1. Why does the present VLSI circuits use MOSFETs instead of BJTs?
Answer

Compared to BJTs, MOSFETs can be made very small as they occupy very small silicon area on IC chip and are relatively simple in terms of manufacturing. Moreover digital and memory ICs can be implemented with circuits that use only MOSFETs i.e. no resistors, diodes, etc.

2. What are the various regions of operation of MOSFET? How are those regions used?
Answer

MOSFET has three regions of operation: the cut-off region, the triode region, and the saturation region.
The cut-off region and the triode region are used to operate as switch. The saturation region is used to operate as amplifier.

3. What is threshold voltage?
Answer

The value of voltage between Gate and Source i.e. V_{GS} at which a sufficient number of mobile electrons accumulate in the channel region to form a conducting channel is called threshold voltage (V_t is positive for NMOS and negative for PMOS).

4. What does it mean "the channel is pinched off"?
Answer

For a MOSFET when V_{GS} is greater than V_t, a channel is induced. As we increase V_{DS} current starts flowing from Drain to Source (triode region). When we further increase V_{DS}, till the voltage between gate and channel at the drain end to become V_t, i.e. $V_{GS} - V_{DS} = V_t$, the channel depth at Drain end decreases almost to zero, and the channel is said to be pinched off. This is where a MOSFET enters saturation region.

5. Explain the three regions of operation of a MOSFET.
Answer

Cut-off region: When $V_{GS} < V_t$, no channel is induced and the MOSFET will be in cut-off region. No current flows.
Triode region: When $V_{GS} \geq V_t$, a channel will be induced and current starts flowing if $V_{DS} > 0$. MOSFET will be in triode region as long as $V_{DS} < V_{GS} - V_t$.
Saturation region: When $V_{GS} \geq V_t$, and $V_{DS} \geq V_{GS} - V_t$, the channel will be in saturation mode, where

the current value saturates. There will be little or no effect on MOSFET when V_{DS} is further increased.

6. What is channel-length modulation?
Answer

In practice, when V_{DS} is further increased beyond saturation point, it does has some effect on the characteristics of the MOSFET. When V_{DS} is increased the channel pinch-off point starts moving away from the Drain and towards the Source. Due to which the effective channel length decreases, and this phenomenon is called as Channel Length Modulation.

7. Explain depletion region.
Answer

When a positive voltage is applied across Gate, it causes the free holes (positive charge) to be repelled from the region of substrate under the Gate (the channel region). When these holes are pushed down the substrate they leave behind a carrier-depletion region.

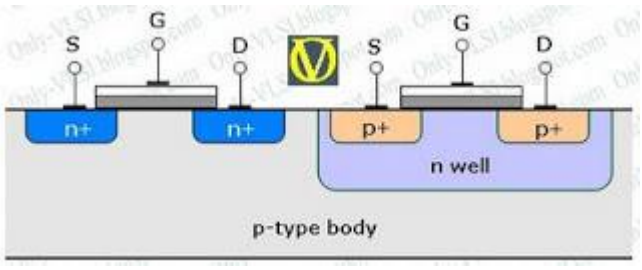
8. What is body effect?
Answer

Usually, in an integrated circuit there will be several MOSFETs and in order to maintain cut-off condition for all MOSFETs the body substrate is connected to the most negative power supply (in case of PMOS most positive power supply). Which causes a reverse bias voltage between source and body that effects the transistor operation, by widening the depletion region. The widened depletion region will result in the reduction of channel depth. To restore the channel depth to its normal depth the VGS has to be increased. This is effectively seen as change in the threshold voltage - V_t. This effect, which is caused by applying some voltage to body is known as body effect.

9. Give various factors on which threshold voltage depends.
Answer

As discussed in the above question, the V_t depends on the voltage connected to the Body terminal. It also depends on the temperature, the magnitude of V_t decreases by about 2mV for every 1^oC rise in temperature.

10. Give the Cross-sectional diagram of the CMOS.
Answer



FPGA: A Field-Programmable Gate Array (FPGA) is a semiconductor device containing programmable logic components called "logic blocks", and programmable interconnects. Logic

blocks can be programmed to perform the function of basic logic gates such as AND, and XOR, or more complex combinational functions such as decoders or mathematical functions.

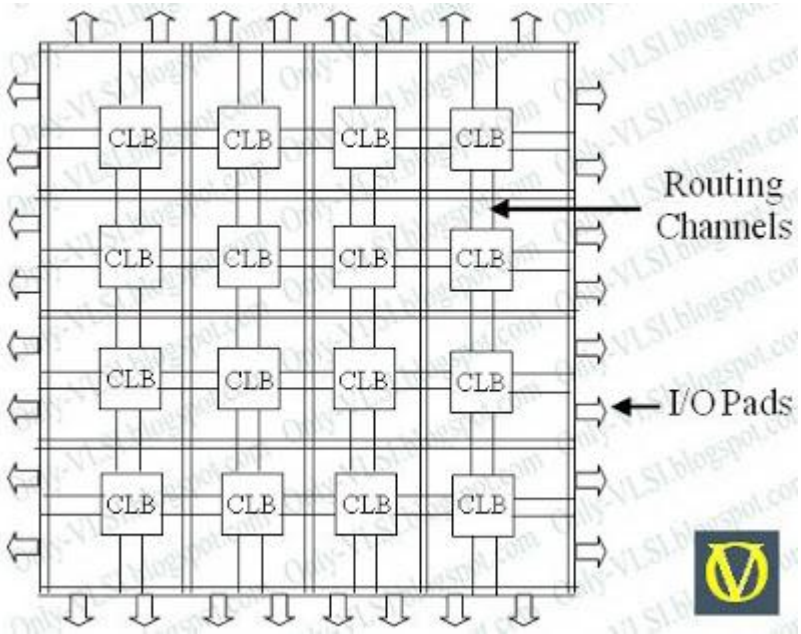
In most FPGAs, the logic blocks also include memory elements, which may be simple flip-flops or more complete blocks of memory.

Applications

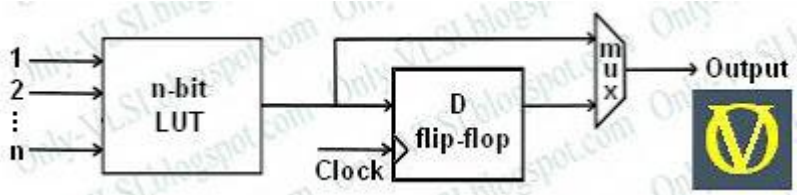
- ASIC prototyping: Due to high cost of ASIC chips, the logic of the application is first verified by dumping HDL code in a FPGA. This helps for faster and cheaper testing. Once the logic is verified then they are made into ASICs.
- Very useful in applications that can make use of the massive parallelism offered by their architecture. Example: code breaking, in particular brute-force attack, of cryptographic algorithms.
- FPGAs are sued for computational kernels such as FFT or Convolution instead of a microprocessor.
- Applications include digital signal processing, software-defined radio, aerospace and defense systems, medical imaging, computer vision, speech recognition, cryptography, bio-informatics, computer hardware emulation and a growing range of other areas.

Architecture

FPGA consists of large number of "configurable logic blocks" (CLBs) and routing channels. Multiple I/O pads may fit into the height of one row or the width of one column in the array. In general all the routing channels have the same width. The block diagram of FPGA architecture is shown below.



CLB: The CLB consists of an n-bit look-up table (LUT), a flip-flop and a 2x1 mux. The value n is manufacturer specific. Increase in n value can increase the performance of a FPGA. Typically n is 4. An n-bit lookup table can be implemented with a multiplexer whose select lines are the inputs of the LUT and whose inputs are constants. An n-bit LUT can encode any n-input Boolean function by modeling such functions as truth tables. This is an efficient way of encoding Boolean logic functions, and LUTs with 4-6 bits of input are in fact the key component of modern FPGAs. The block diagram of a CLB is shown below.



Each CLB has n-inputs and only one input, which can be either the registered or the unregistered LUT output. The output is selected using a 2x1 mux. The LUT output is registered using the flip-flop (generally D flip-flop). The clock is given to the flip-flop, using which the output is registered. In general, high fanout signals like clock signals are routed via special-purpose dedicated routing networks, they and other signals are managed separately.

Routing channels are programmed to connect various CLBs. The connecting done according to the design. The CLBs are connected in such a way that logic of the design is achieved.

FPGA Programming

The design is first coded in HDL (Verilog or VHDL), once the code is validated (simulated and synthesized). During synthesis, typically done using tools like Xilinx ISE, FPGA Advantage, etc, a technology-mapped net list is generated. The net list can then be fitted to the actual FPGA architecture using a process called place-and-route, usually performed by the FPGA company's proprietary place-and-route software. The user will validate the map, place and route results via timing analysis, simulation, and other verification methodologies. Once the design and validation process is complete, the binary file generated is used to (re)configure the FPGA. Once the FPGA is (re)configured, it is tested. If there are any issues or modifications, the original HDL code will be modified and then entire process is repeated, and FPGA is reconfigured.

ASIC: An application-specific integrated circuit (ASIC) is an integrated circuit designed for a particular use, rather than intended for general-purpose use. Processors, RAM, ROM, etc are examples of ASICs.

FPGA vs ASIC

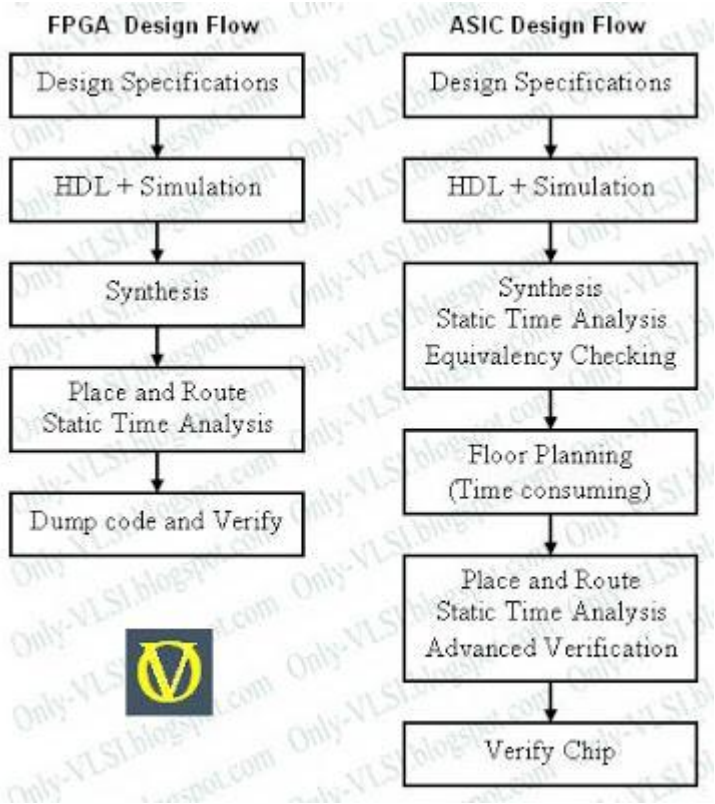
Speed
ASIC rules out FPGA in terms of speed. As ASIC are designed for a specific application they can be optimized to maximum, hence we can have high speed in ASIC designs. ASIC can have hight speed clocks.

Cost
FPGAs are cost effective for small applications. But when it comes to complex and large volume designs (like 32-bit processors) ASIC products are cheaper.

Size/Area
FPGA are contains lots of LUTs, and routing channels which are connected via bit streams(program). As they are made for general purpose and because of re-usability. They are in-general larger designs than corresponding ASIC design. For example, LUT gives you both registered and non-register output, but if we require only non-registered output, then its a waste of having a extra circuitry. In this way ASIC will be smaller in size.

Power
FPGA designs consume more power than ASIC designs. As explained above the unwanted circuitry results wastage of power. FPGA wont allow us to have better power optimization. When it comes to ASIC designs we can optimize them to the fullest.

Time to Market
FPGA designs will till less time, as the design cycle is small when compared to that of ASIC designs. No need of layouts, masks or other back-end processes. Its very simple: Specifications -- HDL + simulations -- Synthesis -- Place and Route (along with static-analysis) -- Dump code onto FPGA and Verify. When it comes to ASIC we have to do floor planning and also advanced verification. The FPGA design flow eliminates the complex and time-consuming floor planning, place and route, timing analysis, and mask / re-spin stages of the project since the design logic is already synthesized to be placed onto an already verified, characterized FPGA device.



Type of Design
ASIC can have mixed-signal designs, or only analog designs. But it is not possible to design them using FPGA chips.

Customization
ASIC has the upper hand when comes to the customization. The device can be fully customized as ASICs will be designed according to a given specification. Just imagine implementing a 32-bit processor on a FPGA!

Prototyping
Because of re-usability of FPGAs, they are used as ASIC prototypes. ASIC design HDL code is first dumped onto a FPGA and tested for accurate results. Once the design is error free then it is taken for further steps. Its clear that FPGA may be needed for designing an ASIC.

Non Recurring Engineering/Expenses
NRE refers to the one-time cost of researching, designing, and testing a new product, which is generally associated with ASICs. No such thing is associated with FPGA. Hence FPGA designs are

cost effective.

Simpler Design Cycle
Due to software that handles much of the routing, placement, and timing, FPGA designs have smaller designed cycle than ASICs.

More Predictable Project Cycle
Due to elimination of potential re-spins, wafer capacities, etc. FPGA designs have better project cycle.

Tools
Tools which are used for FPGA designs are relatively cheaper than ASIC designs.

Re-Usability
A single FPGA can be used for various applications, by simply reprogramming it (dumping new HDL code). By definition ASIC are application specific cannot be reused.